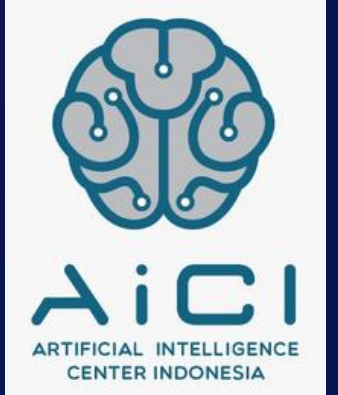




**Kampus
Merdeka**
INDONESIA JAYA



NLP – Topic Modeling



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Topic Modeling

- Input, A document-term matrix. Each topic will consist of a set of words where order doesn't matter, so we're going to start with the bag of words format.
- Gensim, gensim is a Python toolkit built by Radim Řehůřek specifically for topic modeling. We're going to a popular topic modeling technique called Latent Dirichlet Allocation (LDA). We're also going to use nltk for some parts-of-speech-tagging.
- **Output**, Our goal is to find themes across various comedy routines, and see which comedians tend to talk about which themes.

1. Question

2. Data

3. Perform
EDA

4. Techniques

5. Insight



Topic Modeling

Latent Dirichlet Allocation

1. Question

2. Data

3. Perform
EDA

4. Techniques

5. Insight



Topic Modeling

Latent Dirichlet Allocation

Hidden

Type of probability
distribution

1. Question

2. Data

3. Perform
EDA

4. Techniques

5. Insight



Latent Dirichlet Allocation

I like
bananas
and
oranges

100% Topic
A

Frogs
and fish
live in
ponds

100% Topic
B

Kittens
and
puppies
are
fluffy

100% Topic
A

I had a
spinach
and
apple
smoothie

100% Topic
B

My
kitten
loves
kale

60% Topic A
40% Topic B

1. Question

2. Data

3. Perform
EDA

4. Techniques

5. Insight



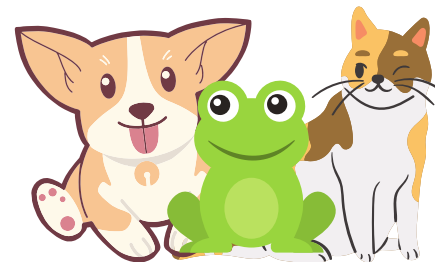
Latent Dirichlet Allocation

Topic A: 40% banana, 30% kale, 10% breakfast...



Food

Topic B: 30% kitten, 20% puppy, 10% frog, 5% cute....



Animals

1. Question

2. Data

3. Perform
EDA

4. Techniques

5. Insight



Latent Dirichlet Allocation

I like
bananas
and
oranges

Food

Frogs
and fish
live in
ponds

Animals

Kittens
and
puppies
are
fluffy

Food

I had a
spinach
and
apple
smoothie

Animals

My
kitten
loves
kale

Food +
Animals

1. Question

2. Data

3. Perform
EDA

4. Techniques

5. Insight



Latent Dirichlet Allocation

Every **document** consists of
a mix of **topics**

I like
bananas
and
oranges

Food

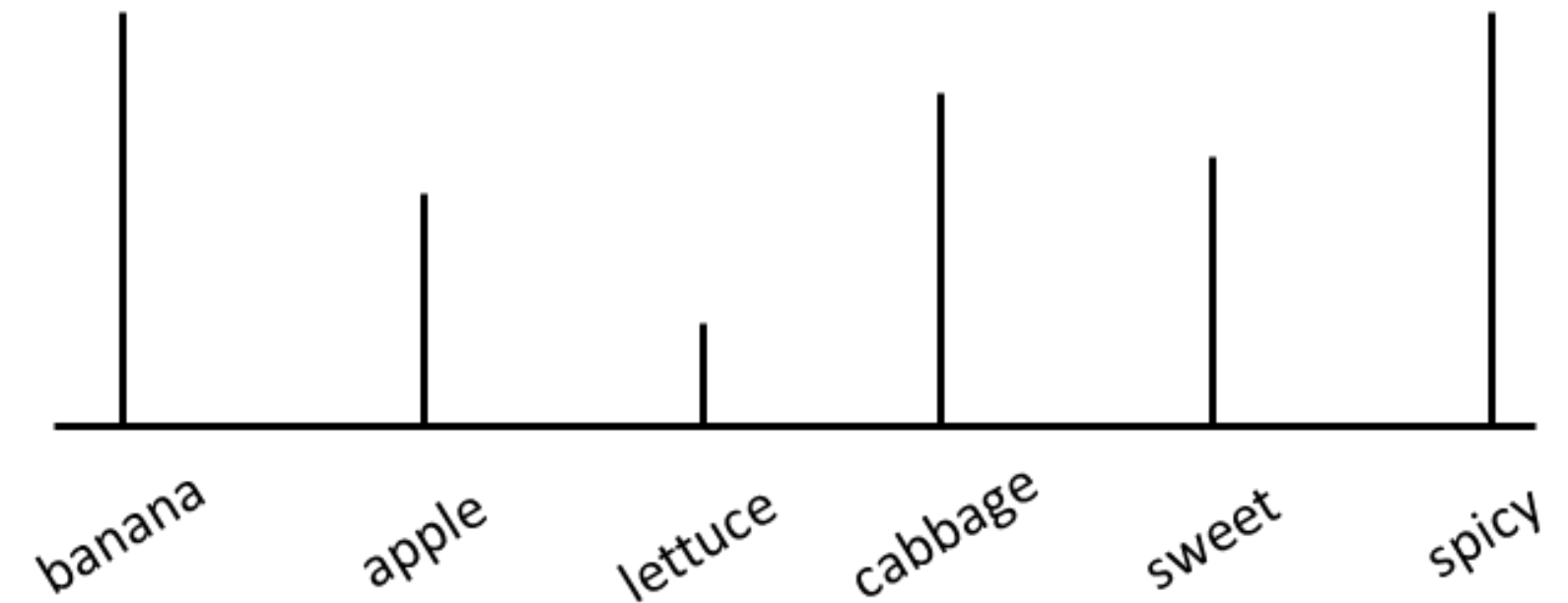
Frogs
and fish
live in
ponds

Animals

My
kitten
loves
kale

Food

Every **topic** consists of
a mix of **words**



Topic: Food

1. Question

2. Data

3. Perform
EDA

4. Techniques

5. Insight



Latent Dirichlet Allocation

How LDA Works

- Goal: you want LDA to learn the topic mix in each document and the word mix in each topic
- Choose the number of topics you think there are in your corpus

Example: $K = 2$

I like
bananas
and
oranges

Doc #1

Topic A:
Food

Topic B:
Animals

1. Question

2. Data

3. Perform
EDA

4. Techniques

5. Insight



Latent Dirichlet Allocation

How LDA Works

- Goal: you want LDA to learn the topic mix in each document and the word mix in each topic
- Choose the number of topics you think there are in your corpus

Example: $K = 2$

- Randomly assign each word in each document to one of 2 topics

Example: The word 'banana' in #Doc1 is randomly assigned to Topic B (animal-like topic)

- Go through every word and its topic assignment in each document. Look at (1) how often the topic occurs in the document and (2) how often the word occurs in the topic overall. Based on this info, assign the word a new topic.?

Example: It looks like (1) animals don't occur often in #Doc1 and (2) 'banana' doesn't occur much in topic B, so the word 'banana' probably should be assigned to topic A instead

I like
bananas
and
oranges

Doc #1

Topic A:
Food

Topic B:
Animals

1. Question

2. Data

3. Perform
EDA

4. Techniques

5. Insight



Latent Dirichlet Allocation

How LDA Works

- Goal: you want LDA to learn the topic mix in each document and the word mix in each topic
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Example: It looks like (1) animals don't occur often in #Doc1 and (2) 'banana' doesn't occur much in topic B, so the word 'banana' probably should be assigned to topic A instead

- Go through multiple iterations of this. Eventually, the topics will start making sense - interpret them.

I like
bananas
and
oranges

Doc #1

Topic A:
Food

Topic B:
Animals

1. Question

2. Data

3. Perform
EDA

4. Techniques

5. Insight



Latent Dirichlet Allocation

How LDA Works

- Goal: you want LDA to learn the topic mix in each document and the word mix in each topic
- Choose the number of topics you think there are in your corpus

Example: $K = 2$

I like
bananas
and
oranges

Doc #1

Topic A:
Food

Topic B:
Animals

Gensim does this part for you



- Go through multiple iterations of this. Eventually, the topics will start making sense - interpret them.

1. Question

2. Data

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EDA

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Latent Dirichlet Allocation

- **Input:** A document-term matrix, number of topics, number of iterations
- **Gensim:** goes through the process of finding the best word distribution for each topic and best topic distribution for each document.
- **Output: The top words in each topic.** It is your job as a human to interpret this and see if the results make sense or not. If not, try altering the parameters – terms in the document-term matrix, number of topics, number of iterations, etc. **Stop when the topics make sense.**
- This is probabilistic approach to topic modelling. There are also matrix factorization techniques for topic modelling such as **Latent Semantic Indexing (LSI)** and **Non-Negative Matrix Factorization (NMF)**

1. Question

2. Data

3. Perform
EDA

4. Techniques

5. Insight



Google Colab Code

1. Question

2. Data

3. Perform
EDA

4. Techniques

5. Insight